

Curriculum Vitae

MAHMOUD MOHAMED YOUSSEF ABD-RABBOU

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PERSONAL DATA

- **Date of Birth:** 1 Jan. 1990.
- **Place of Birth:** Beheira, Egypt.
- **Private Place:** Cairo, Egypt.
- **Nationality:** Egyptian.
- **Passport Number:** A36973579
- **Gender:** Male.
- **Marital status:** Married to Hager El-sharnopy .
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- **Web of Science Researcher ID:** [KVB-5502-2024](#)



RESEARCH METRICS

- h-index: [16] (Scopus), [17] WoS, [18] (Google Scholar).
- Total Citations: [665] (Scopus), [712] WoS, [800] (Google Scholar).
- ResearchGate Score: 905.

EDUCATION

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|-------------------------------------|-----------|------------------------------|
| • Al-Azhar University, Cairo, Egypt | 2018-2020 | Ph.D. (Quantum Information). |
| • Al-Azhar University, Cairo, Egypt | 2014-2018 | M.Sc. (Quantum Optics). |
| • Al-Azhar University, Cairo, Egypt | 2009-2012 | B.Sc. (Applied Mathematics) |

SCIENTIFIC EMPLOYMENTS

- **2024-....** *Postdoctoral* at School of Physics, University of Chinese Academy of Science, Yuquan Road 19A, Beijing, China.
- **2021-....** *Lecturer* at Al-Azhar University, Faculty of Science, Mathematics Department, Cairo, Egypt.
- **2018-2021** *Assistant Lecturer* at Al-Azhar University, Faculty of Science, Mathematics Department, Cairo, Egypt.
- **2014-2018** *Demonstrator of applied mathematics* at Al-Azhar University, Faculty of Science, Mathematics Department, Cairo, Egypt.
- **2013-2014** *Demonstrator* at Modern Academy for Engineering & Technology, Ma'adi, Cairo. Egypt.

SCIENTIFIC ACTIVITIES

- December (2024), **presented two papers** in the 5th International Conference on Data Analytics for Business and Industry, ICDABI, Bahrain.
- November (2024), **presented two papers** in the 6-th International Conference for Mathematics and Its Applications, (ICMA24), EGYPT.
- March (2023), **presented a paper** in the 5-th International on Mathematics and Its Applications, ICMA23, EGYPT.
- Jun (2019), **presented a paper** in the 8-th INTERNATIONAL CONFERENCE ON MODELING SIMULATION AND APPLIED OPTIMIZATION, ICMSAO 2019 Bahrain.
- December (2017), **attended** the International Conference on Mathematics: Trends and Developments (ICMTD17), Ain-Shams University Welcoming House, Cairo, Egypt.
- March (2017), **presented a paper** in the 9-th International Conference for Basic Sciences in Al-Azhar Conference Center (ACC), Nasr city, Cairo, Egypt.
- November (2016), **presented a paper** in One-day Conference for Mathematics and Applications, Al-Azhar University (Assiut Branch), Assiut, Egypt.
- December (2015), **presented a paper** in the International Conference for Mathematics and Applications (ICMA15) in MSA University, 6th of October city, Egypt.

COURSES AND WORKSHOPS

1. Nature research academies" Research Methodology", (2022).
2. Nature research academies" Grant Writing", (2022).
3. Nature research academies" Writing A Research Paper", (2022).
4. Springer Nature Experiments for the EKB: "How to find the protocols and methods you need for your research" (2022).
5. Springer Nature with the EKB "Open Access for Egypt" (2022).
6. CABI with the EKB "How to write professional responses to reviewers!", (2022).

ARABIC COURSES

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- The National Authority for Quality Assurance and Accreditation of Education "Curriculum maps and course descriptions", (2023).
 - The National Authority for Quality Assurance and Accreditation of Education "Self-study preparation", (2023).
 - The National Authority for Quality Assurance and Accreditation of Education "Scientific researches design techniques", (2023).
 - The National Authority for Quality Assurance and Accreditation of Education "Exams and Students' Assessment Systems", (2023).
 - The National Authority for Quality Assurance and Accreditation of Education "Effective presentation skills", (2023).

TECHNICAL SKILLS

I-COMPUTATIONAL & PROGRAMMING:

- Python (NumPy, SciPy, Qiskit), Mathematica, MATLAB.
- Quantum Simulation: QuTiP, Qiskit.
- LaTeX (Advanced).

II-LANGUAGES:

- Arabic: Native.
- English: Fluent (Academic writing & presentation).

PUBLICATIONS

I- ARTICLES IN REFEREED JOURNALS

1. **M. Y. Abd-Rabbou**, A.A. Zahia, A.M. Abdallah, A.A. Gouda, C.-F. Qiao, "Detecting entanglement in high-spin quantum systems via a stacking ensemble of machine learning models" *Scientific Reports*, 16(1), 19181: (2026).
<https://doi.org/10.1038/s41598-026-53789-6>.
2. **M. Y. Abd-Rabbou**, R. Altuijri, R.T. Matoog, A. H. Abdel-Aty" "Floquet-engineered selective sideband control in driven two-qubit systems" *Chaos, Solitons & Fractals*, 211, 118745: (2026).
<https://doi.org/10.1016/j.chaos.2026.118745>.
3. M.A. EL-Sayed, **M. Y. Abd-Rabbou**, S. Esmail "Steady-State Entanglement and Steering in a Hybrid Exciton-Optomechanical System" *Brazilian Journal of Physics*, 56(4), 158: (2026).
<https://doi.org/10.1007/s13538-026-02076-w>.
4. **M. Y. Abd-Rabbou**, A. M. Abdallah, I. Kedim, A.-B.A. Mohamed "A neural surrogate model for efficient statistical analysis of entanglement in light-matter systems" *European Physical Journal Plus*, 141(7):760 (2026).
<https://doi.org/10.1140/epjp/s13360-026-07927-8>.
5. S.K. Singh, A.A. Zahia, J. -X Peng, **M. Y. Abd-Rabbou** "Cavity Magnomechanical Framework for a Highly Efficient Quantum Battery" *Annalen Der Physik*, 538(6), e70225 (2026).
<https://doi.org/10.1002/andp.70225>.
6. **M. Y. Abd-Rabbou**, A. N. Al-Ahmadi, H. Ardah, A. H. Abdel-Aty" Quantum information scrambling and entanglement generation in a Rydberg-dressed atom driven by tunable long-range interactions and quantum fluctuations" *Chaos, Solitons & Fractals*, 207, 117995: (2026).
<https://doi.org/10.1016/j.chaos.2026.117995>.

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7. E.S. Elkhoully, **M. Y. Abd-Rabbou**, S.I. Ali, N. Metally "Sharing a classical string utilizing quantum techniques" *Quantum Information Processing*, 25(6), 190 (2026).
<https://doi.org/10.1007/s11128-026-05213-1>.
 8. A. T. M. Makram-Allah, **M. Y. Abd-Rabbou**, N. Metally "The possibility of fairly mitigating penalties in the quantum Prisoner's Dilemma via classical noise channels" *European Physical Journal Plus*, 141(4), 456 (2026).
<https://doi.org/10.1140/epjp/s13360-026-07650-4>.
 9. S Elghaayda, **M. Y. Abd-Rabbou**, M Mansour" Quantum obesity and steering ellipsoids for fermionic fields in a dilation black hole" *Fortschritte der Physik* , 74 (1), e70068: (2026).
<https://doi.org/10.1002/prop.70068>.
 10. A. A. Zahia, **M. Y. Abd-Rabbou**, E. M. Khalil, S. Al-Awfi" Optimizing quantum battery performance via multi-photon transitions: energy, power, work extraction, and stability" *Journal of Taibah University for Science* , 19 (1), 2565052: (2026).
<https://doi.org/10.1080/16583655.2025.2565052>.
 11. **M.Y.Abd-Rabbou**, C.-F. Qiao" Strengthening the uncertainty and the reverse uncertainty relation limits" *Quantum Information Processing*, 24(12), 387: (2025).
<https://doi.org/10.1007/s11128-025-04992-3>.
 12. Esraa Reda, **M.Y.Abd-Rabbou**, S. I. Ali, E. M. Khalil, and A.-S. F. Obada" Unveiling quantum features of time-dependent electron spins influenced by Zeeman energy and symmetric cross spin" *Journal of Physics B: Atomic, Molecular and Optical Physics*, 58, 205501: (2025).
<https://doi.org/10.1088/1361-6455/adf92b>.
 13. Ahmed A Zahia, **M.Y.Abd-Rabbou**, E. M. Khalil, S. Al-Awfi" Optimizing quantum battery performance via multi-photon transitions: energy, power, work extraction, and stability" *Journal of Taibah University for Science* 19 (1), 2565052 (2025).
<https://doi.org/10.1080/16583655.2025.2565052>
 14. R.I. Mohamed, **M.Y.Abd-Rabbou**, N. Metwally, " Fractional Rashba-induced quantum correlations and their role in achieving steerability in nanowires" *Physica Scripta*, 100(10), 105111: (2025).
<https://doi.org/10.1088/1402-4896/ae0c6c>.
 15. F. Benabdallah, **M.Y.Abd-Rabbou**, M. Daoud, "Non-Markovian Protection and Thermal Fragility of Quantum Resources in a Spin-1/2 Ising–Heisenberg Diamond Chain" *Annalen Der Physik*, 537(10), e00258: (2025).
<https://doi.org/10.1002/andp.202500258>.
 16. A.T.M. Makram-Allah, **M.Y.Abd-Rabbou**, N. Metwally, "Implementing quantum prisoner dilemma via different classes of entangled and separable states" *Modern Physics Letters A*, 40(27), 2550109: (2025).
<https://doi.org/10.1142/S0217732325501093>.
 17. **M.Y.Abd-Rabbou**, H. Allhibi, F. Aljuaydi, A.B.A. Mohamed, "Charging a quantum battery with interacting spin baths in a trio coherent state" *Quantum Information Processing*, 24(8), 245: (2025).
<https://doi.org/10.1007/s11128-025-04854-y>.
 18. **M.Y.Abd-Rabbou**, Ahmed A Zahia, Atta ur Rahman, C. Feng Qiao, " The limits of quantum information scrambling" *Journal of Physics A: Mathematical and Theoretical*, 58(25), 255301: (2025).
<https://doi.org/10.1088/1751-8121/ade1b9>.
 19. S Elghaayda, A Ali, **M.Y.Abd-Rabbou**,M Mansour, S Al-Kuwari "Quantum correlations and metrological advantage among Unruh–DeWitt detectors in de Sitter spacetime" *The European Physical Journal C* , 85 (4), 1-11: (2025).
<https://doi.org/10.1140/epjc/s10052-025-14175-0>.

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20. Ahmed A Zahia, **M.Y.Abd-Rabbou**, A. M. Megahed " Entanglement-driven energy exchange in a two-qubit quantum battery" *Journal of Physics B: Atomic, Molecular and Optical Physics*, 58 (6), 065501: (2025).
<https://doi.org/10.1088/1361-6455/adbc56>.
 21. E. M. Khalil, Ahmed A Zahia, **M.Y.Abd-Rabbou**, "Controlling entanglement and steering in a damped qubit-photon-magnon system under external field influence" *Optical and Quantum Electronics*, 57 (2), 164: (2025).
<https://doi.org/10.1007/s11082-025-08048-9>.
 22. E. S. Elkhoully, **M.Y.Abd-Rabbou**, S. I. Ali, N. Metwally " Quantum kernel as an indicator of secure classical message coded in a quantum string" *Europhysics Letters*, 149 (2), 28001: (2025).
<https://doi.org/10.1209/0295-5075/ada576>.
 23. L. Mashhor, **M.Y.Abd-Rabbou**, Abdel-Azim Abdel-Hady Yaya, M. Abdel-Aty, A.-S. F. Obada" Quantum Otto and Carnot thermal machines powered by two two-level atoms enveloped by QED-cavity" *Indian Journal of Physics*, 1-9 (2025).
<https://doi.org/10.1007/s12648-024-03507-7>.
 24. Fadwa Benabdallah, **M.Y.Abd-Rabbou**, Mohammed Daoud, Saeed Haddadi "Quantum information resources in spin-1 Heisenberg dimer systems" *Quantum Information Processing*, 24, 20: (2025).
<https://doi.org/10.1007/s11128-024-04631-3>
 25. Ahmed A Zahia, **M.Y.Abd-Rabbou**, Ahmed M Megahed" Explicit and implicit quantum correlation dynamics of qubits interacting with thermal baths" *Physica Scripta*, 100 (2), 025101: (2025).
<https://doi.org/10.1088/1402-4896/ada206>.
 26. A. A. Zahia, **M.Y.Abd-Rabbou**, E. M. Khalil, S. Al-Awfi " Quantum battery systems: Analyzing two-mode field charging, qubit interactions, and environmental influences" *Thermal Science*, 28, 6B, 5179-5192: (2024).
<https://doi.org/10.2298/TSCI2406179Z>.
 27. E. M. Khalil, **M.Y.Abd-Rabbou**, H. Alotaibi, R. Althobaiti, "External field-induced squeezing in a four-level atom inside a cavity system" *Thermal Science*, 28(6 Part B):4855-65, (2024).
<https://doi.org/10.2298/TSCI2406855K>.
 28. E. S. Elkhoully, **M.Y.Abd-Rabbou**, S. I. Ali, N. Metwally "Distributing Classical Information via Quantum Secret Sharing Technique" *IEEE Xplore*, (2024).
Published in: 2024 5th International Conference on Data Analytics for Business and Industry (ICDABI)
<https://doi.org/10.1109/ICDABI63787.2024.10800213>.
 29. A.T.M. Makram-Allah, **M.Y.Abd-Rabbou**, N. Metwally" Initial State Time-Dependent Quantum Game" *IEEE Xplore*, (2024).
Published in: 2024 5th International Conference on Data Analytics for Business and Industry (ICDABI)
<https://doi.org/10.1109/ICDABI63787.2024.10800664>.
 30. A.T.M. Makram-Allah, **M.Y.Abd-Rabbou**, N. Metwally" Time dependence of Eisert-Wilkens-Lewenstein quantum game" *Quantum Information Processing*, 23(12), 393: (2024).
<https://doi.org/10.1007/s11128-024-04589-2>.
 31. M. D. Alsulami, **M.Y.Abd-Rabbou**" Quantum Heat Engines with Spin-Chain-Star Systems" *Annalen der Physik*, 536(10), 2400122 (2024).
<https://doi.org/10.1002/andp.202400122>.
 32. **M.Y.Abd-Rabbou**, E.M. Khalil, and Saud Al-Awfi "Quantum Otto machine in Lipkin-Meshkov-Glick model with magnetic field and a symmetric cross interaction" *Optical and Quantum Electronics*, 56(6), 940: (2024).
<https://doi.org/10.1007/s11082-024-06444-1>.
 33. A.S.F. Obada, **M.Y.Abd-Rabbou**, S. Haddadi "Does conditional entropy squeezing indicate normalized entropic uncertainty relation steering?" *Quantum Information Processing*, 23(3), 90: (2024).
<https://doi.org/10.1007/s11128-024-04298-w>.
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34. **M.Y.Abd-Rabbou**, E.M. Khalil, and Saud Al-Awfi "Probing teleported quantum correlations in a two-qubit system inside a coherent field" *Optik*, 296, 171551: (2024).
<https://doi.org/10.1016/j.ijleo.2023.171551>.
 35. Esraa Reda, **M.Y.Abd-Rabbou**, S. I. Ali, and A.-S. F. Obada "Quantum correlations in Heisenberg chain model: Effects of dipolar interaction, symmetric and anti-symmetric interactions" *Modern Physics Letters A*, 2450186: (2024).
<https://doi.org/10.1142/S0217732324501864>.
 36. S. Haddadi, M. A. Yurischev, **M.Y.Abd-Rabbou**, M. Azizi, M. R. Pourkarimi, M. Ghominejad "Quantumness near a Schwarzschild black hole" *European Physical Journal C*, 84(1), 42: (2024).
<https://doi.org/10.1140/epjc/s10052-024-12393-6>.
 37. **M.Y.Abd-Rabbou**, E.M. Khalil, and Saud Al-Awfi "Noise-Based Damping of Chaotic Entanglement in Pulsed Driven Two-Qubit System" *Annalen der Physik*, 536(3), 2300423: (2024).
<https://doi.org/10.1002/andp.202300423>.
 38. **M.Y.Abd-Rabbou**, Atta ur Rahman, Mikhail A. Yurischev, and Saeed Haddadi "Comparative study of quantum Otto and Carnot engines powered by a spin working substance" *Phys. Rev. E*. **108**, (3), 034106: (2023).
<https://doi.org/10.1103/PhysRevE.108.034106>.
 39. Ahmed A. Zahia, **M.Y.Abd-Rabbou**, Ahmed M. Megahed, A.-S. F. Obada "Bidirectional field-steering and atomic steering induced by a magnon mode in a qubit-photon system" *Scientific Reports*. **13**, (1) 14943: (2023).
<https://doi.org/10.1038/s41598-023-41907-7>.
 40. Adel Bandar Alruqi, E.M. Khalil, S. Abdel-Khalek, **M.Y.Abd-Rabbou** "Visualization of the interaction dynamics between a two-level atom and a graphene sheet covered by a laser field" *Alexandria Engineering Journal*. **77**, Pages 309-317: (2023).
<https://doi.org/10.1016/j.aej.2023.06.075>.
 41. Adel Bandar Alruqi, E.M. Khalil, S. Abdel-Khalek, **M.Y.Abd-Rabbou** "Entanglement of a cavity with a three-level atom and a graphene membrane in the presence of a laser field" *Alexandria Engineering Journal*. **75**, Pages 419-428: (2023).
<https://doi.org/10.1016/j.aej.2023.05.070>.
 42. **M.Y.Abd-Rabbou**, M. Shamirzaie, and Salman Khan "Non-classical correlations of accelerated observers interacting with a classical stochastic noise beyond the single mode approximation" *Optical and Quantum Electronics*. **55**, 175: (2023).
<https://doi.org/10.1007/s11082-023-04926-2>.
 43. **M.Y.Abd-Rabbou**, S. S. Hansen "Protecting the fidelity of a pulsed-driven harmonic oscillator" *Optik*. **286**, 170985: (2023).
<https://doi.org/10.1016/j.ijleo.2023.170985>.
 44. M. Youssef, S. I. Ali, **M.Y.Abd-Rabbou**, A. -S. F. Obada "Exploring quantum correlations of two-qubit Heisenberg chain model influenced by magnetic dipole-dipole, magnetic field, and a symmetric cross interaction" *Quantum Information Processing*. **22**, 229: (2023).
<https://doi.org/10.1007/s11128-023-03969-4>.
 45. N. Ibrahim Mohammed, Hanadi M. Abdelsalam, Shaimaa Almalki, **M.Y.Abd-Rabbou**, S. Abdel-Khalek, E. M. Khalil "Witnessing quantum correlations in two coupled quantum dots under intrinsic decoherence" *Alexandria Engineering Journal*. **69**, pp. 521-527: (2023).
<https://doi.org/10.1016/j.aej.2023.01.057>.
 46. Atta Ur Rahman, **M.Y.Abd-Rabbou**, S. Haddadi, Hazrat Ali "Two-Qubit Steerability, Nonlocality, and Average Steered Coherence under Classical Dephasing Channels" *Annalen der Physik*. **535**, (4), 2200523: (2023).
<https://doi.org/10.1002/andp.202200523>.
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47. **M.Y.Abd-Rabbou**, S. I. Ali, N. Metwally "Detraction of decoherence that arises from the acceleration process" *Journal of the Optical Society of America B: Optical Physics*.**40**, (3), pp. 585–593: (2023).
<https://doi.org/10.1364/JOSAB.477867>.
48. Atta Ur Rahman, M. Shamirzaie, **M.Y.Abd-Rabbou** "Bidirectional steering, entanglement and coherence of accelerated qubit–qutrit system with a stochastic noise" *Optik*.**274**, 170543, (2023).
<https://doi.org/10.1016/j.ijleo.2023.170543>.
49. Maged Faihan Alotaibi, E.M. Khalil, **M.Y.Abd-Rabbou**, Marin Marin "The Classicality and Quantumness of the Driven Qubit-Photon-Magnon System" *Mathematics*.**10**, (23), 4458, (2022).
<https://doi.org/10.3390/math10234458>.
50. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada, "Decoherence and quantum steering of accelerated qubit-qutrit system" *Quantum Information Processing*.**21**, 363, (2022).
<https://doi.org/10.1007/s11128-022-03711-6>.
51. E.M. Khalil, **M.Y.Abd-Rabbou**, "Robustness of a teleported state influenced by dipole interaction and magnetic field under intrinsic decoherence" *Optik*.**267**, 169703, (2022).
<https://doi.org/10.1016/j.ijleo.2022.169703>.
52. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada, "Improving the bidirectional steerability between two accelerated partners via filtering process" *Modern Physics Letters A*.**37**, 2250143, (2022).
<https://doi.org/10.1142/S0217732322501437>.
53. Atta Ur Rahman, **M.Y.Abd-Rabbou**, S.M. Zangi, Muhammad Javed" Entropic uncertainty and quantum correlations dynamics in a system of two qutrits exposed to local noisy channels" *Physica Scripta* **97**, 105101, (2022),
<https://doi.org/10.1088/1402-4896/ac8bb1>
54. **M.Y.Abd-Rabbou**, E.M. Khalil, and Fadhel Almalki" Dynamical Features of Isolated Two-and Three-Level Atoms Interacting with a Cavity Field " *Advances in Mathematical Physics* **2022**, 5258035, (2022),
<https://doi.org/10.1155/2022/5258035>
55. **M.Y.Abd-Rabbou**, E.M. Khalil," Dense Coding and Quantum Memory Assisted Entropic Uncertainty Relations in a Two-Qubit State Influenced by Dipole and Symmetric Cross Interactions" *Annalen der Physik* **534**, 2200204, (2022),
<https://doi.org/10.1002/andp.202200204>
56. **M.Y.Abd-Rabbou**, S. Khan, M. Shamirzaie," Quantum Fisher information and quantum coherence of an entangled bipartite state interacting with a common classical environment in accelerating frames" *Quantum Inf Process* **21**, 218, (2022),
<https://doi.org/10.1007/s11128-022-03560-3>
57. Maged Faihan Alotaibi, E.M. Khalil, and **M.Y.Abd-Rabbou**, Mohamed Omri, "Dynamics of an atomic system associated with a cavity-optomechanical system " *Results in Physics* **37**, 105540, (2022),
<https://doi.org/10.1016/j.rinp.2022.105540>
58. E.M. Khalil, Hanaa Abu-Zinadah , and **M.Y.Abd-Rabbou**, "Influence of an External Classical Field on a $\diamond$ Four-Level Atom Inside a Quantized Field". *Symmetry* **41** (4),811 (2022)
<https://doi.org/10.3390/sym14040811>
59. Mohamed Omri, **M.Y.Abd-Rabbou**, E.M. Khalil, S Abdel-Khalek," Thermal information and teleportation in two-qutrit Heisenberg XX chain model" *Alexandria Engineering Journal* **61** (10),8335 (2022)
<https://doi.org/10.1016/j.aej.2022.01.051>
60. **M.Y.Abd-Rabbou**, S. I. Ali, and M.M.A. Ahmed, "Enhancing the information of nonlinear SU (1, 1) quantum systems interacting with a two-level atom", *Optical Quantum Electronic* **54**, 548, (2022)
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61. Maged Faihan Alotaibi, E.M. Khalil, S Abdel-Khalek, **M.Y.Abd-Rabbou**, Mohamed Omri, "Effects of the vibrating graphene membrane and the driven classical field on an atomic system coupled to a cavity field". *Results in Physics* **31**, 105012, (2021),
<https://doi.org/10.1016/j.rinp.2021.105012>
 62. A.-B.A.Mohamed, E. M. Khalil, **M.Y.Abd-Rabbou**, "Nonclassical atomic system dynamics time-dependently interacts with finite entangled pair coherent parametric converter cavity fields". *Optical and Quantum Electronics* **53**, 344, (2021),
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 63. A.-B.A.Mohamed, E. M. Khalil, **M.Y.Abd-Rabbou**, "Quantum effects induced by two classical fields in a coherent cavity field containing two two-level atoms". *Physica E: Low-dimensional Systems and Nanostructures* (134) 114839, (2021),
<https://doi.org/10.1016/j.physe.2021.114839>
 64. D.A.M. Abo-Kahla, **M.Y.Abd-Rabbou**, N. Metwally, "The orthogonality speed of two-qubit state interacts locally with spin chain in the presence of Dzyaloshinsky–Moriya interaction" *Laser Physics Letters* **18** (4), 045203 (2021)
<https://doi.org/10.1088/1612-202X/abe7de>.
 65. **M.Y.Abd-Rabbou**, E. M. Khalil, Sayed Abdel-Khalek, A Al-Barakaty, Hanaa Abu-Zinadah "Quantum Fisher Information of a Teleported State in Heisenberg XYZ Chain With Magnetic Field and Kaplan–Shekhtman–Entin-Wohlman–Aharony Interaction" *IEEE Access* (9), 51325 - 51331, (2021).
<https://doi.org/10.1109/ACCESS.2021.3069163>.
 66. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada, "Wigner Distribution of Accelerated Tripartite W-state" *Optik*.**208**. (2020):163921
<https://doi.org/10.1016/j.ijleo.2019.163921>.
 67. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada, " Suppressing the information losses of accelerated qubit–qutrit system" *International Journal of Quantum Information*, **17**, 4, 1950032. (2019)
<https://doi.org/10.1142/S0219749919500321>.
 68. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada," Wigner function of noisy accelerated two-qubit system" *Quantum Information Processing*,**18**. 12. 367. (2019)
<https://doi.org/10.1007/s11128-019-2480-6>.
 69. **M.Y.Abd-Rabbou**, N. Metwally, M. M. A. Ahmed, A.-S. F. Obada, "Restraining the decoherence of accelerated qubit–qutrit system via local Markovian channels" *Physica Scripta*, **94**,10 105103. (2019)
<https://doi.org/10.1088/1402-4896/ab2427>.
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